

# Darren (Sizuo) Liu

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## Education

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### Sichuan University

*BE in Software Engineering*

- GPA: 3.1/4.0

*Sept 2017 – May 2021*

### Binghamton University

*MS in Computer Science*

- GPA: 3.45/4.0

*Aug 2020 – Dec 2022*

### Metis

*Associate Degree in Data Science and Machine Learning*

*Jun 2022 – Jan 2023*

### Emory University

*PhD in Computer Science and Informatics*

- GPA: 3.82/4.0
- **Track:** Biomedical Informatics

*Aug 2024 – May 2028  
(Expected)*

## Experience

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### Software Engineer Intern

*Chinasoft International*

- Full-stack web development for their employee management website.
- In charge of requirement analysis and database design.

*Chongqing, China  
July 2020 – Aug 2020*

### Part-time Recording Engineer, Songwriter, Producer

*Silence Music*

- In charge of chorus recording and vocal direction.
- In charge of composing and producing.
- Co-produced Yichuan Wang's album "Stop Daydreaming".
- Produced my albums "Murderer" and "DARREN".

*Chengdu, China  
Mar 2020 – Aug 2024*

### Information Analyst

*Emory University*

- Built ModelMeetsData, a full-stack React platform that hosts machine learning models so users can benchmark them on their own datasets and model owners can track performance across datasets.
- Led an LLM evaluation study assessing model performance in tagging clinical concepts detected by MetaMap from nursing notes.
- Built CRCWeb, a full-stack React Native app that delivers educational materials to colorectal cancer patients for managing psychoneurological symptoms (fatigue, depression, sleep disturbance) and collects phone sensor data to analyze activity and recommend daily routines.
- Developed co-occurring token and LLM-output evaluation panels for annotators to perform structured annotations.
- Led a study evaluating large language and vision models on pattern recognition for real-world physiological signals.
- Built a React Native app that visualizes patients' daily activities against their goals, automatically generates new goals from activity data, and rewards goal completion.

*Atlanta, GA  
Mar 2023 – Jun 2024*

### Data Scientist Intern

*Nihon Kohden America*

- Developed and deployed a web-based real-time alarm analytics dashboard (previously deployed at Children's

*Irvine, CA  
May 2025 – Aug 2025*

Healthcare of Atlanta) into the Nihon Kohden Digital Health Platform (NKDHP).

- Developed and deployed a receiver application that pulls data from a SQL server (Bedmaster), converts it into the NKDHP format, and sends it to the Kafka feed.
- Tested and deployed the end-to-end system.

## Projects

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### NutriCamp

[nutricamp.ai](https://nutricamp.ai) 

#### *AI Diet Management Application*

- Built a full-stack diet management application with an embedded multimodal LLM agent that extracts micronutrient information directly from food images, letting users track intake without manual logging.
- Generates personalized dietary suggestions from logged intake, building on the DietAI24 nutrition-estimation framework and the NutriOrion multi-agent intervention system.
- Deployed to production and publicly available at [nutricamp.ai](https://nutricamp.ai) .

### CanBeWell

[canbewell.help](https://canbewell.help) 

#### *Mobile Health Intervention Platform*

- Built a mobile health platform that lets clinicians design and deploy personalized physical-activity and nutrition interventions through a no-code interface, with content changes reaching patients in real time without app rebuilds or store reviews.
- Integrated NutriScan Lite, a camera-based feature that analyzes meals and returns nutritional breakdowns, alongside combined movement and nourishment guidance in a single app.
- Aggregated Fitbit wearable data, ecological momentary assessments, and smartphone sensor data into one longitudinal record for clinical review; publicly available at [canbewell.help](https://canbewell.help) .

## Publications

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### **(In Review) NutriOrion: A Hierarchical Multi-Agent Framework for Personalized Nutrition Intervention Grounded in Clinical Guidelines**

Feb 2026

Wu, J, Yan, R, Luo, H, **Liu, D.**, Wang, M, Townsend, K, Hartwig, L, Milketinas, D, Hu, X, Yang, C

<https://arxiv.org/pdf/2602.18650> 

### **DietAI24 As A Framework for Comprehensive Nutrition Estimation Using Multimodal Large Language Models**

Nov 2025

Yan, R, Luo, H, Lu, J, **Liu, D.**, Posluszny, H, Dhaliwal, M, MacLeod, J, Qin, Y, Yang, C, Hartman, T, Hu, X

<https://www.nature.com/articles/s43856-025-01159-0> 

### **Leveraging Artificial Intelligence for Digital Symptom Management in Oncology: The Development of CRCWeb**

July 2025

**Liu, D.**, Lin, Y, Yan, R, Wang, Z, Bold, D, Hu, X

<https://cancer.jmir.org/2025/1/e68516> 

### **Prediction of cardiac arrest in the pediatric cardiac intensive care unit: A time-series machine learning approach**

Apr 2025

Lu, J., Brown, S., Wu, Y., Dong, K., Bold, D, **Liu, D.**, Grunwell, J., Hu, X.

<https://www.sciencedirect.com/science/article/abs/pii/S0883944124004453> 

### **Evaluation of Large Language Models in Tailoring Educational Content for Cancer Survivors and Their Caregivers: Quality Analysis**

Feb 2025

**Liu, D.**, Hu, X, Xiao, C, Bai, J, Lee, S, ..., Lin, Y

<https://cancer.jmir.org/2025/1/e67914> 

### **1444: Using Machine Learning to Predict Cardiac Arrest in the Pediatric Cardiac Intensive Care Unit**

Jan 2025

Brown, S. Grunwell, J. Wu, Y. Dong, K. Bold, D. **Liu, D.**, Fundora, M. Hu, X. Lu, J.

<https://journals.lww.com/ccmjournal/> 

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Perceptions and Needs for A Technology-Based Dyadic Intervention on Symptom Management Among Patients with Colorectal Cancer and Their Caregivers: A Qualitative Study</b> Epari, A., Kim, K., Xiao, C., Porter, L. S., Alese, O. B., Northouse, L., <i>Liu, D.</i> , Bold, D., Graetz, I., Lin, Y<br><a href="https://journals.lww.com/cancernursingonline/">https://journals.lww.com/cancernursingonline/</a> | Sep 2024 |
| <b>Evaluation of General Large Language Models in Understanding Clinical Concepts Extracted from Adult Critical Care Electronic Health Record Notes</b> <i>Liu, D.</i> , Ding, C., Bold, D., Bouvier, M., Lu, J., Jabaley, CS. ..., Hu, X.<br><a href="https://arxiv.org/abs/2401.13588">arxiv.org/abs/2401.13588</a>                                                                                              | Jan 2024 |

## Conferences

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| <b>NutriScan: An AI-Powered, Image-Based Research Tool for Low-Burden Dietary Assessment</b><br><a href="https://nutrition.org/meeting/">https://nutrition.org/meeting/</a>                                                                                                                                                                                     | July 2026<br>NUTRITION    |
| <b>Enabling Scalable Predictive Monitoring and Alarm Analytics via a Real-Time Platform for Processing Continuous Cardiorespiratory Monitoring Data</b><br><a href="https://amia.secure-platform.com/symposium/gallery/rounds/82021/details/20357">https://amia.secure-platform.com/symposium/gallery/rounds/82021/details/20357</a>                            | November 2025<br>AMIA     |
| <b>Enabling Scalable Live Alarm Analytics: AlarmX</b><br><a href="https://bhi.embs.org/2025/">https://bhi.embs.org/2025/</a>                                                                                                                                                                                                                                    | October 2025<br>IEEE BHI  |
| <b>DietAI24: A Novel LLM-Based Dietary Assessment Software From Food Images</b><br><a href="https://cdn.nutrition.org/article/S2475-2991(25)02366-2/fulltext">https://cdn.nutrition.org/article/S2475-2991(25)02366-2/fulltext</a>                                                                                                                              | May 2025<br>Nutrition 9   |
| <b>Empowering Customized Data Annotation: Dynamic Forms</b><br><a href="https://bridge2ai.org/">https://bridge2ai.org/</a>                                                                                                                                                                                                                                      | May 2025<br>Bridge2AI/NIH |
| <b>Systematic Evaluation of General Large Language Models for Contextually Assessed Semantic Concepts From Unstructured Critical Care Data</b><br><a href="https://www.atsjournals.org/doi/pdf/10.1164/ajrccm-conference.2024.209.1_MeetingAbstracts.A5060">https://www.atsjournals.org/doi/pdf/10.1164/ajrccm-conference.2024.209.1_MeetingAbstracts.A5060</a> | May 2024<br>ATS           |
| <b>Do General Large Language and Vision Models Possess Pattern Recognition Abilities for Real-World Physiological Signals?</b><br><a href="https://2024.mcbios.com/program">https://2024.mcbios.com/program</a>                                                                                                                                                 | March 2024<br>MCBIOS      |
| <b>Using Large Language Models to Tag Clinical Concepts Extracted from Nursing Notes</b><br><a href="https://bhi.embs.org/2023/wp-content/uploads/sites/97/2023/10/IEEE-BHI-2023-Program.v4.pdf">https://bhi.embs.org/2023/wp-content/uploads/sites/97/2023/10/IEEE-BHI-2023-Program.v4.pdf</a>                                                                 | October 2023<br>IEEE BHI  |

## Skills

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**Programming Languages:** Python, JavaScript, HTML, CSS, SQL, R, Java, C, C++

**Web & Mobile:** React, React Native, Node.JS, Express.JS, Spring Boot, Flask, MySQL, SQLite, MongoDB

**ML & Data:** PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, Hugging Face, LLM Agents, Machine Learning, Reinforcement Learning

**Tools & Infrastructure:** Git, Docker, Kafka, REST APIs